

Palindrome Queries

Time limit: 1.00 s

Memory limit: 512 MB

You are given a string that consists of n characters between a–z. The positions of the string are indexed $1, 2, \dots, n$.

Your task is to process m operations of the following types:

1. Change the character at position k to x
2. Check if the substring from position a to position b is a palindrome

Input

The first input line has two integers n and m : the length of the string and the number of operations.

The next line has a string that consists of n characters.

Finally, there are m lines that describe the operations. Each line is of the form " $1\ k\ x$ " or " $2\ a\ b$ ".

Output

For each operation 2, print YES if the substring is a palindrome and NO otherwise.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq k \leq n$
- $1 \leq a \leq b \leq n$

Example

Input	Output
7 5 aybabbtu 2 3 5 1 3 x 2 3 5 1 5 x 2 3 5	YES NO YES